Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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2nd Year 1st Semester (Hons)

Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 03, 04

Experiment Title: Simplification of the Boolean Logic expression:
$Y_1 = AB + (B+C)A + (B+C)B$
Basic Gate operations using NAND gate

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Experiment No: 3

Experiment Name:

Simplification of the boolean circuit

$Y_1 = AB + (B+e)A + (B+e)B$

Objectives:

- To simplify the given boolean expression
- To understand the practical significance of boolean simplification in reducing circuit complexity

Theory:

simplifying the boolean equation:

$Y_1 = AB + (B+e)A + (B+e)B$

$= AB + AB + Ae + BB + Be$ [Distributive Law]

$= AB + Ae + B + Be$ [Idempotent Law]

$= B(A+1+e) + Ae$

$= B + Ae$ [Absorption Law]

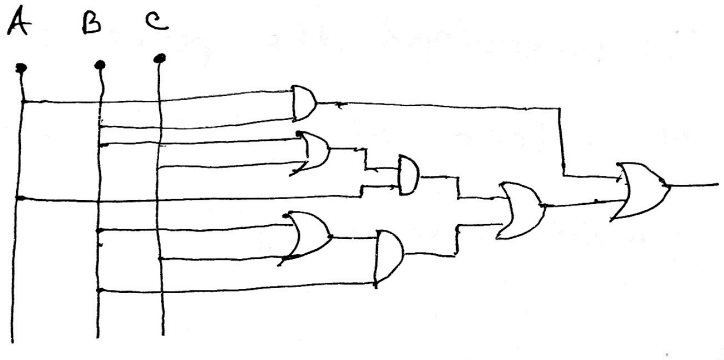
Required Components:

Digital trainer board, bread boards, wires

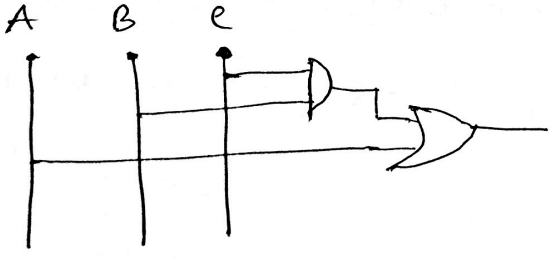
IC 7408, IC 7432.

Circuit Diagram:

Circuit of the expression:



Circuit of the simplified ~~Circuit~~ Expression



Procedure:

- Using IC 7408 and IC 7432 connect the wires to make the circuit of the given expression.
- Then simplify the equation.
- Make another circuit of it.
- connect both output to the LED.
- Verify the output from the truth table

Truth Table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Result:

Both the main and simplified expressions matched with the truth table.

Analysis:

- The simplified form B+AC is minimal and requires fewer logic gates.
- Original circuit requires 5 gates where the simplified only two.

~~Experiment No: 04~~

Experiment Name: Implementing Basic logic gates using NAND Gate.

Objectives:

- To use NAND as an universal gate.
- To understand the basic gates operations

Theory:

1. ~~NAND~~ AND operation using NAND gates

$$A.B = \overline{\overline{A.B}.\overline{A.B}}$$

2. OR operation using NAND gates

$$A+B = \overline{\overline{A.A}.\overline{B.B}}$$

3. NOT operation using NAND gates.

$$\overline{A} = \overline{A.A}$$

4. NOR operation using NAND gates

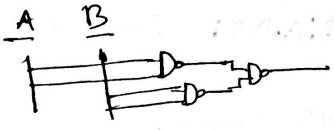
$$\overline{A+B} = \overline{\overline{A} * \overline{B}}$$

Required Components:

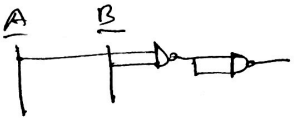
Digital trainer board, wires, NAND Gate (IC7400)

Circuit Diagram:

OR AND:



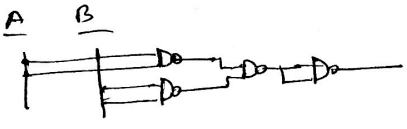
OR



NOT



NOR



Procedure:

- Connect the circuits according to the diagram.
- Check LED outputs.

Result:

All the gate operations worked correctly

Discussion:

It was the experiment of showing the NAND gate as a universal gates through the Basis AND, OR, NOT, NOR operations.